

Problem

The transfer function of a circuit is $H(\omega) = \frac{10}{j\omega + 2}$

If the input signal to the circuit is $v_s(t) = e^{-4t}u(t)$ V, find the output signal. Assume all initial conditions are zero.

Step-by-step solution

Step 1 of 4

Consider the transfer function of a circuit, $H(\omega) = \frac{10}{j\omega + 2}$

Consider the input to the circuit, $v_s(t) = e^{-4t}u(t)$ V

Write the expression for output response, $Y(\omega)$.

$$Y(\omega) = H(\omega)X(\omega) \dots\dots (1)$$

Here,

$H(\omega)$ represents transfer function

$Y(\omega)$ represents output response

$X(\omega)$ represents input excitation, which is the Fourier transform of $v_s(t)$

Step 2 of 4

Calculate the Fourier transform of $v_s(t)$.

$$X(\omega) = \mathcal{F}[e^{-4t}u(t)]$$

Apply the Fourier transform pair $e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a + j\omega}$ by substituting a value as 4.

$$X(\omega) = \frac{1}{j\omega + 4}$$

Substitute $\frac{1}{j\omega + 4}$ for $X(\omega)$ and $\frac{10}{j\omega + 2}$ for $H(\omega)$ in equation (1).

$$\begin{aligned} Y(\omega) &= \left[\frac{10}{j\omega + 2} \right] \left[\frac{1}{j\omega + 4} \right] \\ &= \frac{10}{(j\omega + 2)(j\omega + 4)} \quad \text{Here } A \text{ and } B \text{ are constants.} \\ &= \frac{A}{(j\omega + 2)} + \frac{B}{(j\omega + 4)} \end{aligned}$$

Step 3 of 4

Calculate the constant value A using the residue method.

$$\begin{aligned} A &= (j\omega + 2)Y(\omega)\Big|_{j\omega=-2} \\ &= (j\omega + 2)\left(\frac{10}{(j\omega + 2)(j\omega + 4)}\right)\Big|_{j\omega=-2} \\ &= \left(\frac{10}{(-2 + 4)}\right) \\ &= 5 \end{aligned}$$

Calculate the constant value B using the residue method.

$$\begin{aligned} B &= (j\omega + 4)Y(\omega)\Big|_{j\omega=-4} \\ &= (j\omega + 4)\left(\frac{10}{(j\omega + 2)(j\omega + 4)}\right)\Big|_{j\omega=-4} \\ &= \left(\frac{10}{(-4 + 2)}\right) \\ &= -5 \end{aligned}$$

Step 4 of 4

Substitute obtained A and B values in equation, $Y(\omega)$.

$$Y(\omega) = \frac{5}{(j\omega + 2)} + \frac{-5}{(j\omega + 4)}$$

Apply the inverse Fourier transform to find output signal, $y(t)$.

$$\begin{aligned} y(t) &= \mathcal{F}^{-1}(Y(\omega)) \\ &= \mathcal{F}^{-1}\left[\frac{5}{(j\omega + 2)} + \frac{-5}{(j\omega + 4)}\right] \\ &= \mathcal{F}^{-1}\left(\frac{5}{(j\omega + 2)}\right) - \mathcal{F}^{-1}\left(\frac{5}{(j\omega + 4)}\right) \quad \left(\because e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a + j\omega}\right) \\ &= 5e^{-2t}u(t) - 5e^{-4t}u(t) \\ &= 5(e^{-2t} - e^{-4t})u(t) \end{aligned}$$

Thus, the output signal, $y(t)$ is $\boxed{5(e^{-2t} - e^{-4t})u(t)}$.